Homework 3

Due: Apr. 13th

Turn in your homework in class

Rules:

- Please work on your own. Discussion is permissible, but extremely similar submissions will be judged as plagiarism!
- Please show all intermediate steps: a correct solution without an explanation will get zero credit.
- Please submit **on time**. No late submission will be accepted.
- Please prepare your submission in English only. No Chinese submission will be accepted.
- Please retain **3** decimal places if rounding is required without special requirements for the question,

1 The Source-Free RC Circuit

[6 points] The switch in **Figure 1** has been closed for a long time and opens at $t = 0$.

(a) Find the initial energy stored in the 2 mF capacitor;

(b) Find $v_o(t)$ for $t > 0$.

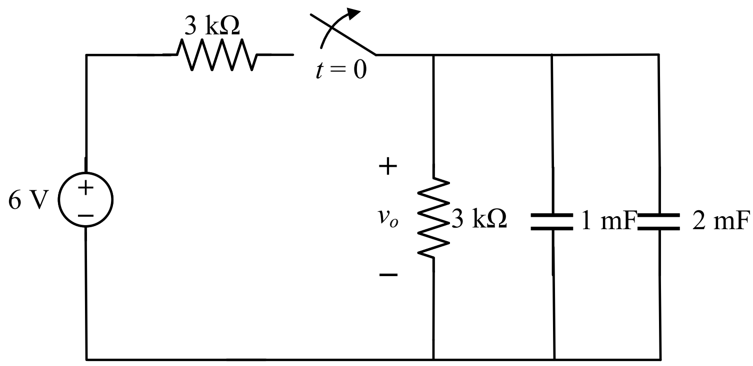

Figure 1

2 The Source-Free *RL* Circuit

[10 points] The switch in **Figure 2** has been in position a for a long time. At $t = 0$, the switch moves instantaneously to position b.

(a) Find the initial energy stored in the inductor;

(b) Find $v_o(t)$ for $t > 0$.

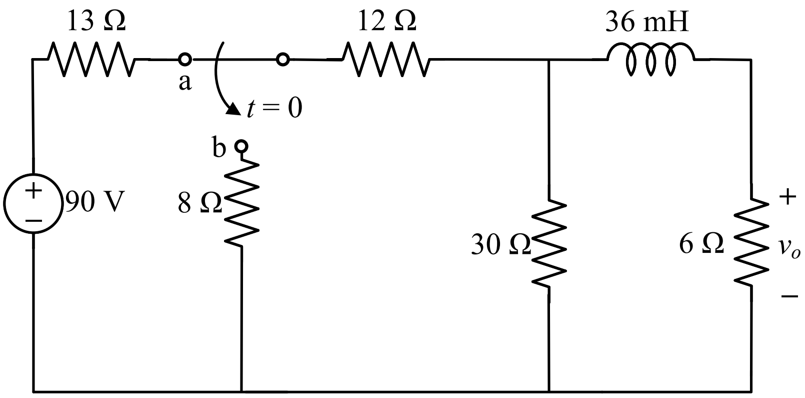

Figure 2

3 Step Response of an RC Circuit

[12 points] Consider the circuit in **Figure 3**. The switch has been closed for a long time and opens at $t = 0$. Find $i(t)$ for $t > 0$.

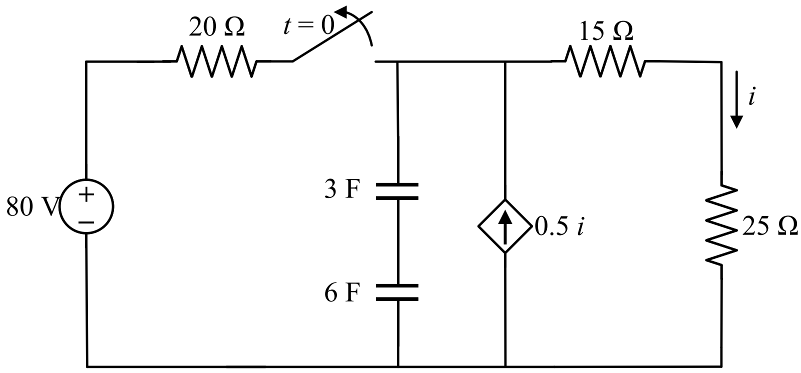

Figure 3

4 Step Response of an RC Circuit

[15 points] The switch in the circuit seen in **Figure 4** has been in position a for a long time. At $t = 0$, the switch moves instantaneously to position b. For $t > 0$, find $v_o(t)$ and $i_o(t)$.

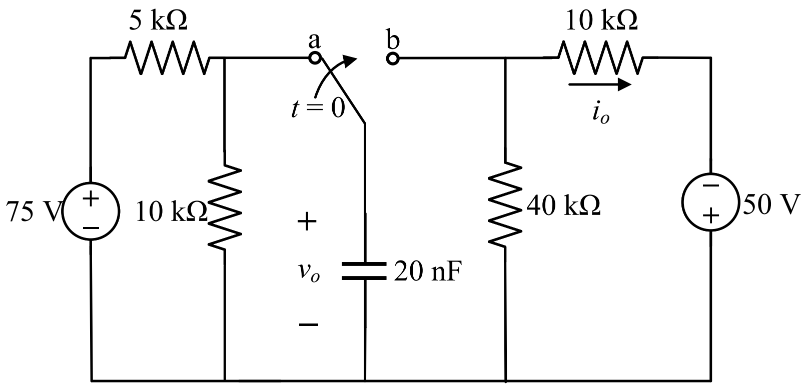

Figure 4

5 Step Response of an *RL* Circuit

[12 points] The switch in the circuit shown in **Figure 5** has been in position a for a long time, and at $t = 0$ it moves to position b. Find $v(t)$ for $t > 0$.

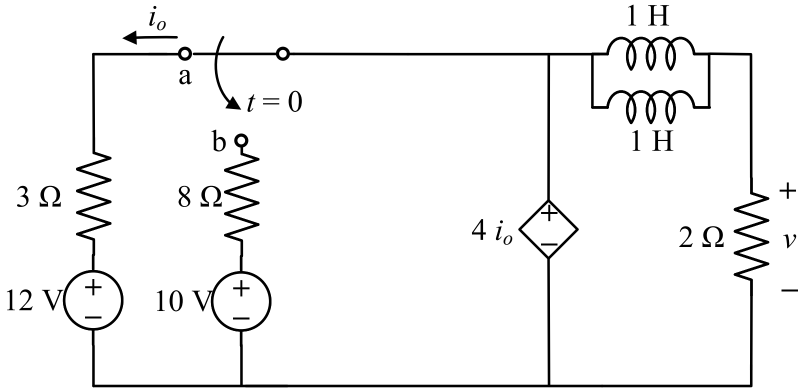

Figure 5

6 Step Response of an *RL* Circuit

[15 points] Assume that the switch in the circuit shown in **Figure 6** has been in position b for a long time, and at $t = 0$ it moves to position a. Find $v(t)$ and $i(t)$ for $t > 0$.

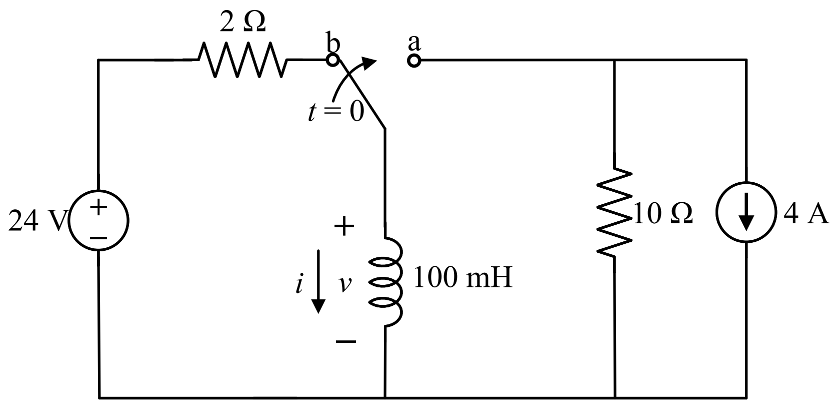

Figure 6

7 The First-Order Op Amp Circuit

[15 points] For the op amp circuit in **Figure 7**, the operational amplifier is ideal and operates in its linear region. Given that $v(0) = 2$ V, find $v_o(t)$ for $t > 0$.

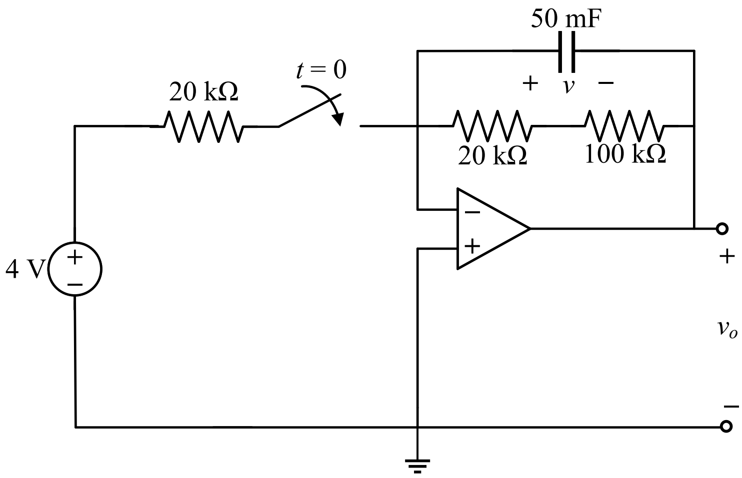

Figure 7

8 The First-Order Op Amp Circuits With Any Excitation

[15 points] In the circuit of **Figure 8**, the operational amplifier is ideal and operates in its linear region. Find $v_o(t)$ and $i_o(t)$ for $t > 0$, given that $v(0) = 1$ V. Consider the following cases:

(a) $v_s(t) = 4\sin t u(t)$;

(b) $v_s(t) = 3tu(t)$;

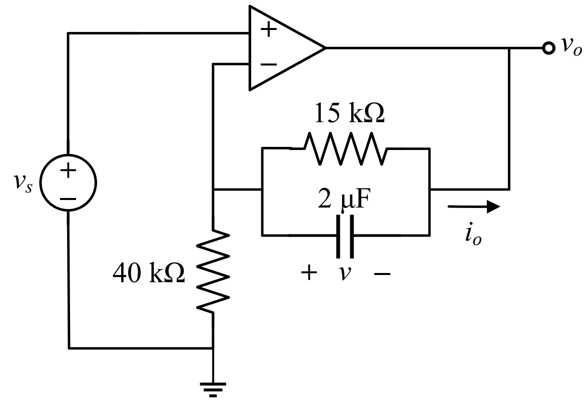

Figure 8